

Impact of metastable defect structures on carrier recombination in solar cells[†]

Seán R. Kavanagh,^{*a,b} David O. Scanlon,^a Aron Walsh^{a,c} and Christoph Freysoldt^d

The efficiency of a solar cell is often limited by electron-hole recombination mediated by defect states within the band gap of the photovoltaic (PV) semiconductor. The Shockley-Read-Hall (SRH) model considers a static trap that can successively capture electrons and holes. In reality however, true trap levels vary with both the defect charge state and local structure. Here we consider the role of metastable structural configurations in capturing electrons and holes, taking the tellurium interstitial in CdTe as an illustrative example. Consideration of the defect dynamics, and symmetry-breaking, changes the qualitative behaviour and activates new pathways for carrier capture. Our results reveal the potential importance of metastable defect structures in non-radiative recombination, in particular for semiconductors with anharmonic/ionic-covalent bonding, multinary compositions, low crystal symmetries or highly-mobile defects.

Submitted to Faraday Discussions, to be presented at the RSC Faraday Discussion on 'Emerging inorganic materials in thin-film photovoltaics' in Bath, UK, July 2022.

Introduction

The search for new thin-film photovoltaic (PV) materials must balance advantageous properties (such as high absorbance, a band gap matching the targeted light spectrum, easy separation of electrons and holes, etc.) with the susceptibility to performance-limiting loss mechanisms. Among these, defect-mediated non-radiative recombination represents the dominant loss mechanism in emerging inorganic PV technologies.^{1,2} Identifying the 'killer' defect traps and hence potential synthesis and processing strategies to avoid their effects is crucial to mitigating losses and achieving high performance. For instance, the major efficiency improvements of the last 5 years for single-junction perovskite solar cells can be directly attributed to effective trap management.³

Point defects, such as vacancies, interstitials and antisites, facilitate carrier recombination by breaking the periodicity of the solid and thus introducing new electronic states within the semiconducting band gap. By successively capturing electrons and holes through non-radiative (vibrational) processes, these defect levels allow excited charge carriers to recombine across the gap, reducing the open-circuit voltage in solar cells and the quantum efficiencies of light-emitting diodes for example.¹ Recent advances in both theory and computation have rendered possible the explicit prediction of defect recombination activity,⁴⁻⁹ allowing for the *ab initio* identification of such 'killer' traps. The synergistic combination of theoretical methods and experimental characterisation has allowed for rapid improvements in photoconversion efficiencies for emerging inorganic PV materials.^{1,10}

Point defects can exhibit complex potential energy surfaces (PESs), with multiple local minima.^{9,11-14} As such, their atomic structures often do not reside solely in their global equilibrium configuration, but may access alternative metastable structures via thermal, photo or electronic conversion. The possibility of structural transformation at defect sites has several potential impacts for technological applications — for example, low energy barriers between structures could lead to coherence corruption and a breakdown in performance for defects used in quantum computing and single-photon emission. In particular, carrier capture rates at defects are sensitively dependent on the atomic structure, varying by over 15 orders of magnitude

^a Department of Chemistry & Thomas Young Centre, University College London, 20 Gordon Street, London WC1H 0AJ, UK

^b Department of Materials & Thomas Young Centre, Imperial College London, Exhibition Road, London SW7 2AZ, UK

^c Department of Materials Science and Engineering, Yonsei University, Seoul 03722, Republic of Korea

^d Max-Planck-Institut für Eisenforschung GmbH, Max-Planck-Str. 1, 40237 Düsseldorf, Germany

* sean.kavanagh.19@ucl.ac.uk

[†] Electronic Supplementary Information (ESI) appended.

with different geometries for the same nominal defect species,^{9,15} and so the presence of these alternative configurations can introduce alternative pathways which significantly alter the overall recombination kinetics. Indeed, this phenomenon has recently been reported in the literature, with metastable defects accelerating the electron-hole recombination process and transforming benign defect centres into harmful traps.^{9,15–17}

A complete analysis of the possible mechanisms and resulting impact of metastable configurations on non-radiative recombination in semiconductors is lacking. In this work, we first provide a general analysis of the potential recombination pathways introduced by metastable defect structures. We discuss the conditions for this behaviour to occur, the anticipated trends and the factors governing the consequences for material properties. We highlight tellurium interstitials in CdTe as illustrative exemplars of the potential complexity and importance of metastable configurations to carrier recombination, exhibiting a novel thermal-excitation recombination cycle which has not previously been reported in the literature, to the knowledge of the authors. Finally, we discuss the important implications of these findings to the broader field of photovoltaic materials research.

Metastable Defect Dynamics

There are several mechanisms by which the presence of metastable structures can influence the overall defect behaviour in solids. Regarding non-radiative carrier recombination, the structural transition pathways introduced by metastable configurations on the defect potential energy surface can act to accelerate or retard this process, or may not affect it at all. Fig. 1 illustrates these possible transitions using a simplified graph network for the case of a single metastable configuration D^* that is both accessible and significantly influential on the overall capture behaviour.

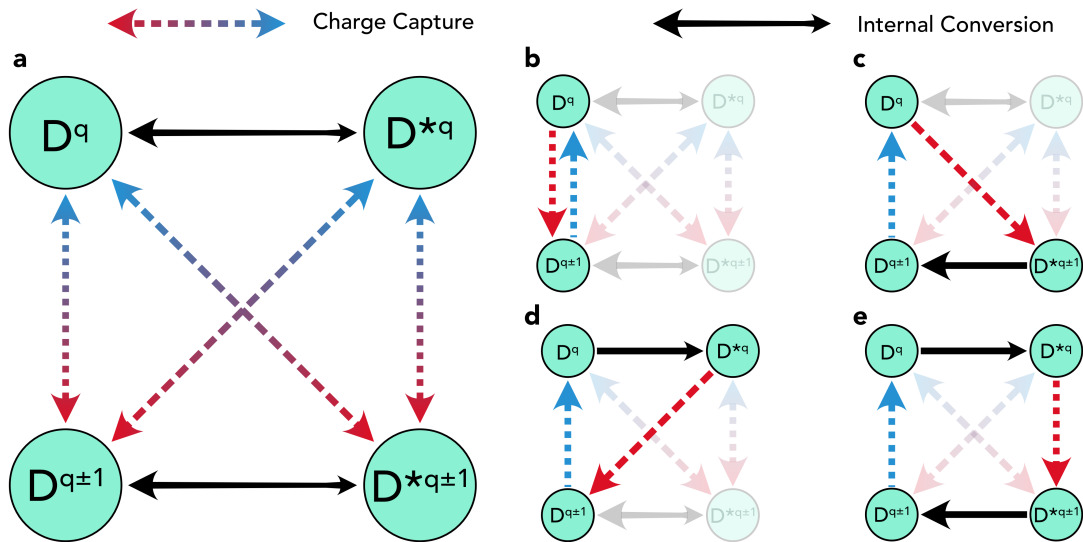


Fig. 1 a. Graph network illustration of the potential structural transitions involving charge capture (red/blue) or internal conversion (black) for a defect D in charge state q . A single metastable configuration D^{*q} is assumed to be accessible for each charge state. b. Pathway for a typical 2-step Shockley-Read-Hall recombination process. c. Pathway for a 3-step recombination process involving *capture* into a metastable defect structure, followed by internal relaxation and charge capture back to the initial state. d. Pathway for a 3-step recombination process involving *internal excitation* to a metastable structure, followed by electron and hole charge capture. e. Pathway for a 4-step recombination process involving *internal excitation* to a metastable structure (D^{*q}), charge capture to the $q \pm 1$ metastable structure ($D^{*q\pm 1}$), *internal relaxation* and charge capture back to the initial configuration (D^q)

The impact of these metastable defects depends on the competing charge capture rates (dashed red/blue arrows), the relative energies of and transition barriers *between* ground and metastable configurations (i.e. their accessibility) (solid black arrows) and the defect energy level position in the host band gap (Eq. (1)).

Each of the charge capture transitions can be fast or slow, with no *necessary* correlation between opposing directions. That said, the recurrence of harmonic PESs means that often a fast electron capture rate implies slow hole capture in the reverse direction, and vice versa.^{1,9,18} This is typically the result of the defect level lying close to the valence or conduction band, often yielding strong/weak interaction with the near/distant band edge and thus fast/slow carrier capture. This is the intuitive rationale behind the expectation that deep midgap defect levels will act as efficient recombination centres.^{1,15,18,19} In many cases however, this trend is violated as a result of anharmonicity in the defect PES, often arising due to symmetry-breaking in the defect structure.^{7,9,20} The prevalence of this phenomenon in semiconductors is becoming increasingly better understood, owing to advances in experimental characterisation and computational modelling techniques.^{1,9,11,21-23} On the other hand, there is a clear and definite correlation between the opposing internal conversion directions, indicated by the relative sizing of the black arrows in Fig. 1. The transition energy barrier from ground to higher energy metastable configuration must be larger than the reverse, hence $k_{D \rightarrow D^*} < k_{D^* \rightarrow D}$ where k is the transition rate constant (i.e. it is always easier to thermally relax than to excite).

Transitions to higher charge states ($q \pm 2$) have not been included here as the behaviour is mostly equivalent and, regardless, for recombination to occur, a closed path must be established (as in Fig. 1b-e), returning the defect to the initial configuration. That said, two ways in which $q \pm 2$ defects *could* influence the overall recombination rate here is (1) by introducing rapid capture into, but slow capture out of, these extremal charge states, thus acting to inhibit recombination as defects get ‘stuck’ in inert charge states,^{1,15,20} or (2) the rare scenario where $D^{q \pm 2}$ introduces an alternative low-barrier pathway between $D^{q \pm 1}$ and $D^{*q \pm 1}$ which does not exist otherwise, potentially accelerating the recombination process (i.e. introducing an additional path along the lower rim of the graph network in Fig. 1a).

Fig. 1c serves as an illustrative example of the potential acceleration or deceleration of the recombination cycle by metastable structures, with the rate of the internal relaxation (black arrow) from $D^{*q \pm 1}$ to $D^{q \pm 1}$ being the deciding factor. Assuming rapid capture transitions for $q \rightarrow q^* \pm 1$ (red) and $q \pm 1 \rightarrow q$ (blue), a fast internal relaxation means that $D^{*q \pm 1}$ acts as a reaction intermediate, introducing a rapid pathway for the recombination cycle to proceed. If, however, internal conversion is extremely slow and $D^{*q \pm 1}$ does not readily relax to $D^{q \pm 1}$, several alternative pathways may emerge. The long-lived metastable configuration could undergo charge capture and return back to the original D^q , completing the e - h recombination cycle without involving $D^{q \pm 1}$. Such behaviour would increase/decrease the overall recombination rate if the rate-limiting step for $D^q \rightleftharpoons D^{*q \pm 1}$ was faster/slower than that of $D^q \rightleftharpoons D^{q \pm 1}$. Instead, if the defect level is sufficiently close to a band edge $D^{*q \pm 1}$ could readily emit the captured charge, acting as a shallow defect level with no effect on the recombination rate. Alternatively, if $D^{*q \pm 1}$ has both slow internal relaxation *and* slow charge capture, defects could become ‘stuck’ in this inert state and in fact greatly diminish the overall recombination activity.

In the above discussion, only non-radiative (phonon emission) pathways are considered. Under certain conditions, both charge capture and internal conversion may also proceed radiatively through photon emission. A detailed consideration of such behaviour is beyond the scope of this report, though we note that it is non-radiative capture which dominates efficiency losses in PV semiconductors.^{1,4,24}

Capture Into Metastable Defects

In recent years, several papers have highlighted the decisive role that metastable defects can play in charge-carrier capture kinetics.^{9,15-17} In each of these studies, it is the situation described in Fig. 1c that is reported, where the defect D^q captures an electron or hole by transforming to a metastable configuration of the $q \pm 1$ defect ($D^{*q \pm 1}$). This metastable species then relaxes to the ground-state $D^{q \pm 1}$ through an internal conversion reaction, whether a spin-flip de-excitation¹⁵ or (more commonly) structural/vibrational relaxation,^{16,17} or both.⁹ In each case, this alternative pathway is found to proceed more rapidly than the standard $D^q \rightleftharpoons D^{q \pm 1}$ charge capture process (Fig. 1b), thus enhancing the overall recombination rate and reconciling theoretical recombination models with experimental observations.

emplified by the temperature-dependent ‘reduced energy’ E_r :^{34–36}

$$N_D = gN_s \exp\left(\frac{-\Delta H}{k_B T}\right) = N_s \exp\left(\frac{-E_r}{k_B T}\right) \quad (12)$$

$$E_r = \Delta H - k_B T \ln(g) \quad (13)$$

where N_D is the defect concentration, g is the site degeneracy, N_s is the concentration of lattice sites and ΔH is the defect formation enthalpy. The ‘reduced energies’ for Te_i are given in Table 1. The small reduced energy difference $E_r(300\text{K}) = 18\text{meV}$ between ground and metastable configurations for Te_i^{*+1} corresponds to an equilibrium concentration ratio of only $[\text{Te}_i^{*+1}]/[\text{Te}_i^{+1}] = \exp\frac{-\Delta E_r}{k_B T} = 0.5$ at room temperature, while for Te_i^0 the greater degeneracy of the ground-state disfavours the metastable configuration. Notably, such entropic effects become even more significant at elevated temperatures (for example during crystal growth and annealing where defects are initially introduced — strongly affecting the resulting self-consistent Fermi level) and for cases with larger degeneracy differences (potentially over an order of magnitude). This phenomenon is another important consequence of symmetry-breaking and metastable configurations for point defects in semiconductors, which can significantly impact bulk defect concentrations and thus Fermi level position, in addition to the effects on non-radiative recombination discussed here.

Table 1 Point group symmetries, site, spin and total (configurational) degeneracies (g_{Site} , g_{Spin} and g_{Total}), relative formation enthalpies (ΔH) and room temperature ‘reduced energies’ ($E_r(T = 300\text{K})$) for tellurium interstitials $\text{Te}_i^{(*)0/+1}$ in CdTe

Defect	Point Symmetry	g_{Site}	g_{Spin}	g_{Total}	ΔH	$E_r(T = 300\text{K})$
—	—	—	—	—	meV	meV
Te_i^0	C_2	12	1	12	0	0
Te_i^{*0}	C_{2v}	6	1	6	127	145
Te_i^{+1}	C_{2v}	6	2	12	0	0
Te_i^{*+1}	C_s	12	2	24	36	18

Te_i in CdTe; Charge Capture. To determine the non-radiative electron/hole capture rates of the Te_i defect levels, we employed the one-dimensional configuration coordinate model of Alkauskas *et al.*⁴ as implemented in the `CarrierCapture.jl` package.⁵ The potential energy surfaces (PESs) for Te_i along the structural paths (configuration coordinates Q) between neutral and positive configurations are shown in Fig. 5.

By calculating the electron-phonon coupling ($W_{if} = \langle \Psi_i | \partial \hat{H} / \partial Q | \Psi_f \rangle$) and solving the 1D Schrödinger equation to determine the effective vibrational wavefunctions for the calculated energy surfaces, the non-radiative recombination activity of the corresponding defect transition levels (TLs) may be computed, with the results for Te_i provided in Table 2 and Fig. 6. The relationship between the calculated PESs and resultant capture rates can be intuitively understood through the classical energy barriers to charge capture, denoted by $\Delta E_{n/p}$ in Fig. 5. A small capture barrier $\Delta E_{n/p}$ and/or close overlap of the defect PESs near the equilibrium position typically yields fast carrier capture, while large barriers and/or significant separation of PESs results in slow capture rates.

Inspecting the defect PESs in Fig. 5 and corresponding capture rates in Table 2 & Fig. 6, we firstly note that all four potential transition pathways exhibit fast *hole* capture; $\text{Te}_i^{(*)0} + h^+ \rightarrow \text{Te}_i^{(*)+1}$, with close overlap of the PESs about the upper orange $\text{Te}_i^{(*)0}$ minimum in each case. This behaviour is consistent with the conventional rationale of fast hole capture for defect levels located near the VBM, as is the case here (Fig. 4). The large hole capture cross-sections $\sigma_p \sim 10^{-15}\text{cm}^2$ — of similar magnitude to its atomic cross-section $\sigma_{\text{Te}} = \pi(a_{\text{Te}})^2 \simeq 1.5 \times 10^{-15}\text{cm}^2$ — classify Te_i as a ‘giant’ hole trap.^{24,37,38} On the other hand, we witness negligible *electron* capture due to high barriers and low PES overlap for the $(+/0)$ and $(+/0^*)$ transition levels (Fig. 5a,b), with tiny capture cross-sections $\sigma_n \sim 10^{-26}\text{cm}^2$. It is only when Te_i^{*+1} is involved, as in

corresponding band-edge sets the energy window within which the metastable structure must lie in order to affect the electron-hole recombination process (as demonstrated in Fig. S1).

S1.2 Charge Capture From Metastable Defects; $D^{*q} \rightarrow D^{q\pm 1}$

In the case of charge capture *from* metastable structures, the same equations from the previous section hold, just with a reversal of the sign of the ΔE term:

$$\epsilon(q^*/q \pm 1) = \epsilon(q/q \pm 1) \pm \Delta E \quad (\text{S20})$$

$$e^- \text{ capture} \implies \epsilon(q^*/q - 1) = \epsilon(q/q - 1) - \Delta E \quad (\text{S21})$$

$$h^+ \text{ capture} \implies \epsilon(q^*/q + 1) = \epsilon(q/q + 1) + \Delta E \quad (\text{S22})$$

Hence for charge capture *from* a metastable structure, $D^{*q} \rightarrow D^{q\pm 1}$, the transition level will move an energy ΔE *further from* the corresponding band-edge, assuming a transition level $\epsilon(q/q \pm 1)$ initially located within the bandgap.

Likewise, we obtain the following reversed constraints on the relative energy of the metastable structure, to impact carrier recombination:

$$0 < \epsilon(q^*/q \pm 1) < E_g \quad (\text{S23})$$

$$0 < \epsilon(q/q \pm 1) \pm \Delta E < E_g \quad (\text{S24})$$

$$e^- \text{ capture} \implies \Delta E < \epsilon(q/q - 1) \quad (\text{S25})$$

$$h^+ \text{ capture} \implies \Delta E < E_g - \epsilon(q/q + 1) \quad (\text{S26})$$

S1.3 Charge Capture Between Two Metastable Defect Structures; $D^{*q} \longleftrightarrow D^{*q\pm 1}$

As demonstrated in Fig. S1, the positions of defect charge transition levels involving metastable configurations for both charge states are dictated by the balance of the relative energies of both metastable structures $D^{*q(\pm 1)}$ with respect to their ground-state counterparts $D^{q(\pm 1)}$, such that:

$$\epsilon(q^*/q^* \pm 1) = \epsilon(q/q \pm 1) \pm \Delta E_q \mp \Delta E_{q\pm 1} \quad (\text{S27})$$

with the same constraint that $\epsilon(q^*/q^* \pm 1)$ must lie within the bandgap in order for recombination to be energetically permitted.

S2 Te_i in CdTe: Nudged Elastic Band (NEB) Calculations

Fig. S2 shows the calculated minimum energy paths between the groundstate and metastable structures for Te_i^0 and Te_i^{+1} . As mentioned in the main text, in both cases the metastable structures represent the intermediate state in the diffusion mechanism of Te_i in CdTe. Thus the energy barriers to structural transition also represent the activation energies for interstitial diffusion of tellurium in CdTe.

Using the HSE(34.5%)+SOC hybrid DFT functional, we calculate transition energy barriers of 0.13 eV and 0.08 eV respectively for $\text{Te}_i^0 \rightarrow \text{Te}_i^{*0}$ and $\text{Te}_i^{+1} \rightarrow \text{Te}_i^{*+1}$. We note that Yang *et al.*, Ma *et al.* both found similar barriers of 0.09 eV for the $\text{Te}_i^0 \rightarrow \text{Te}_i^{*0}$ transition, with semi-local (GGA) and hybrid (HSE06) DFT respectively, and no local stability about Te_i^{*0} . On the other hand, Selvaraj *et al.* calculated a barrier of 0.24 eV using local (LDA) DFT, with a 0.03 eV window of stability for the intermediate state.

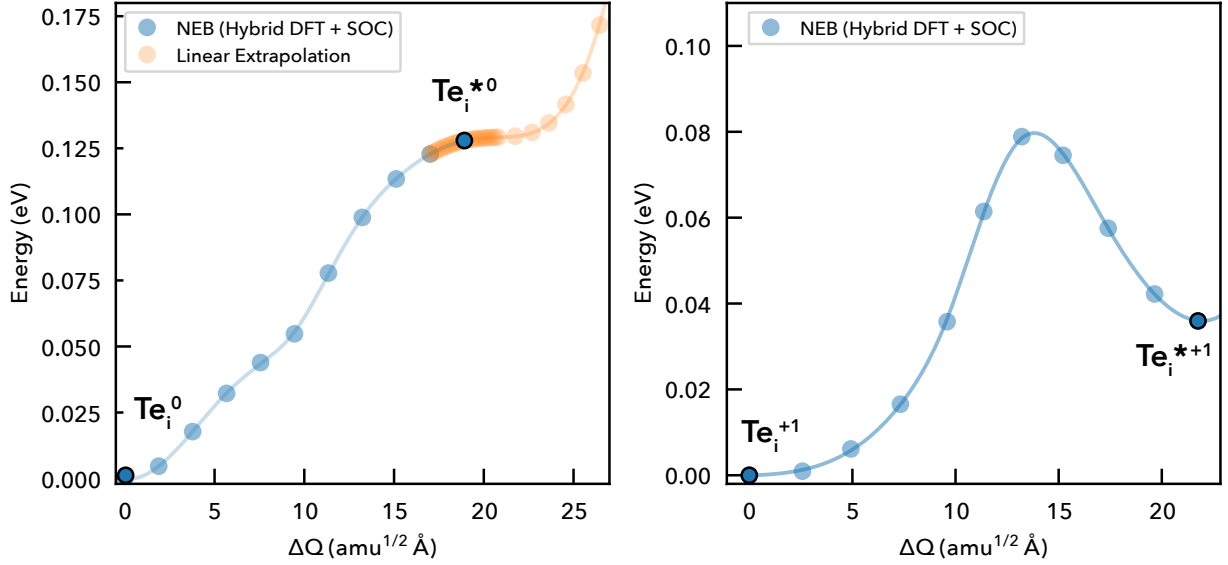


Fig. S2 Minimum energy paths along the potential energy surface between groundstate and metastable structures for **(left)** Te_i^0 and **(right)** Te_i^{+1} , calculated using the Nudged Elastic Band (NEB) method.^{S53} Filled circles represent calculated data points and the solid line is a spline fit. X-axis given in units of mass-weighted displacement.

S3 Te_i in CdTe: Transition State Theory

To estimate the rate of internal structural transformation for point defects, we can invoke Transition State Theory,^{S54} which gives the rate of reaction k as:

$$k = \nu g \exp\left(-\frac{\Delta E}{k_B T}\right) \quad (\text{S28})$$

where ν is the effective (vibrational) attempt frequency, g is the ratio of the degeneracies of the final and initial states and ΔE is the activation energy barrier.

Using the calculated energy barrier of 0.08 eV and effective vibrational attempt frequency $\nu = 0.40$ THz (from parabolic fitting of the Te_i^{+1} local minimum shown in Fig. S2), the $\text{Te}_i^{+1} \rightarrow \text{Te}_i^{*+1}$ transition is estimated to proceed at a rate:

$$k_{+/+*} = (4 \times 10^{11})(4) \exp\left(-\frac{0.08}{(0.0259)(300)}\right) = 7.6 \times 10^{10} \text{ s}^{-1} \text{ @ } T = 300 \text{ K}$$

S3.1 Internal Conversion vs Charge Capture Rates

To estimate the transition barrier height at which internal conversion and charge capture would have comparable speeds, we equate their reaction rates:

$$k = \nu g \exp\left(-\frac{\Delta E}{k_B T}\right) = \sigma_n \nu_{th} n \quad (\text{S29})$$

where σ_n is the capture cross-section, ν_{th} is the carrier thermal velocity and n is the minority carrier concentration (minority carrier capture often represents the rate-limiting step in the electron-hole recombination cycle). This is then rearranged to:

$$\exp\left(-\frac{\Delta E}{k_B T}\right) = \frac{\sigma_n \nu_{th} n}{\nu g} \quad (\text{S30})$$

$$\Delta E = -k_B T \ln \frac{\sigma_n \nu_{th} n}{\nu g} \quad (\text{S31})$$